

Cerebral Blood Flow as a Potential Biomarker of Mood Disturbance in Idiopathic Generalized Epilepsy

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Adapted from a peer-review publication, published on Epilepsia Open 2025.
Prepared as a writing sample.

Depression affects 20–30% of people with IGE but is not routinely screened for in neurology care

- IGE is 15–20% of all epilepsies
- 20–30% of people with IGE have comorbid depression
- The link runs both ways: depression raises the risk of later epilepsy, and epilepsy raises the risk of later depression

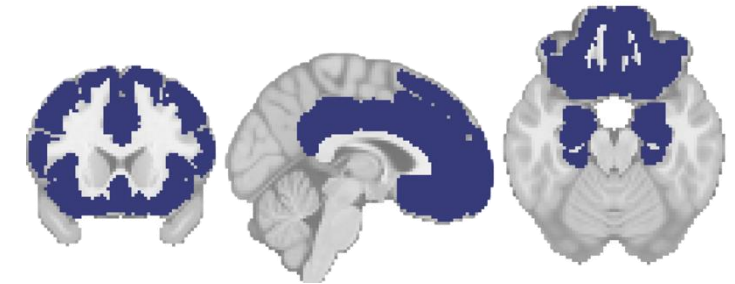
CBF changes appear in both depression and IGE, but had not been linked to mood in IGE

- In young adults at familial risk, higher amygdala CBF predicted the first onset of depression
- In IGE with absence seizures, postictal CBF was increased in temporal and parahippocampal regions
- In IGE, mood has been linked to cortical excitability, but not yet to CBF

23 adults with IGE and 19 healthy controls completed 3T MRI with pulsed arterial spin labeling.

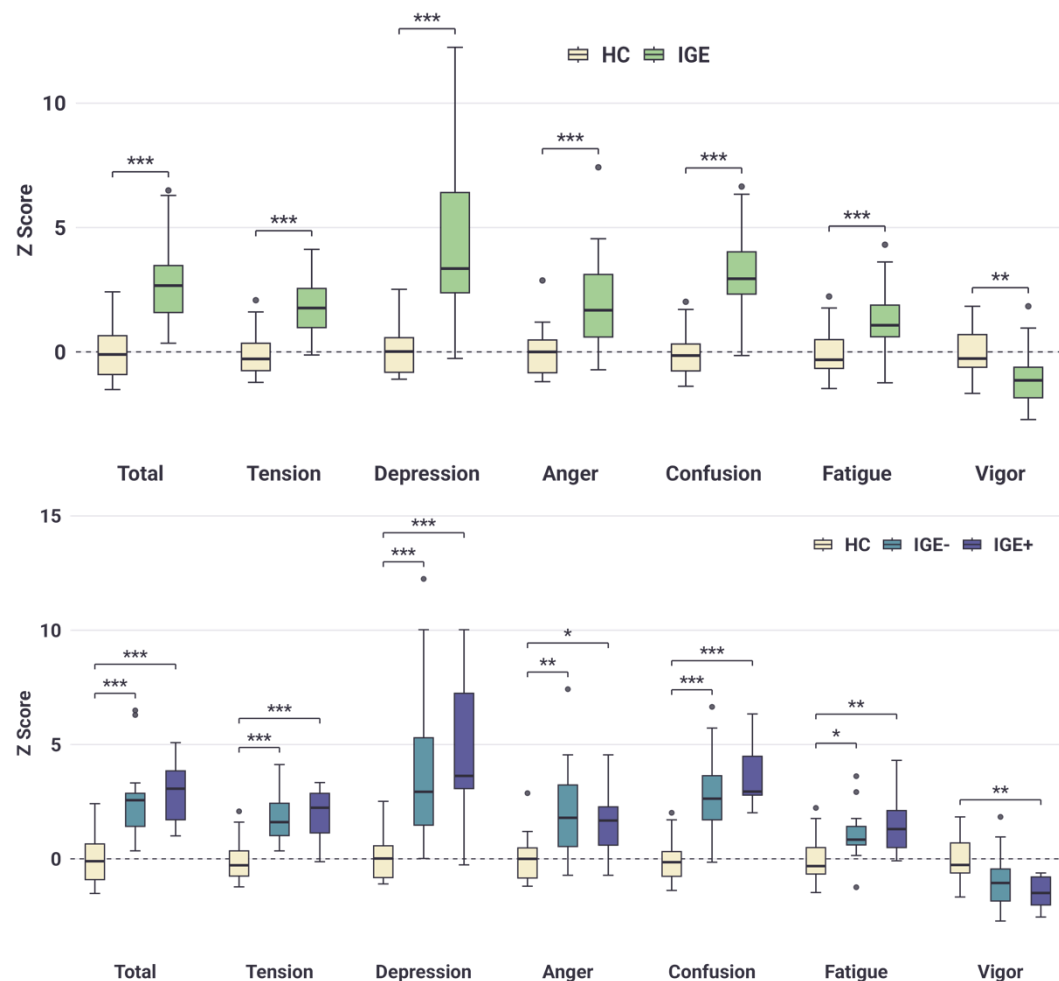
	HC	All	IGE	
			IGE-	IGE+
N	19	23	12	11
Age, years	29.47 (6.90)	31.30 (7.42)	31.17 (9.29)	31.45 (5.13)
Sex, female	9 (47.4%)	13 (56.5%)	8 (66.7%)	5 (45.5%)
Clinical variables				
Age of epilepsy onset, years	-	13.61 (3.98)	12.13 (3.08)	11.95 (4.32)
Duration of epilepsy, years	-	17.70 (8.08)	16.04 (9.33)	19.50 (6.39)
Number of anti- seizure medications	-	1.91 (1.28)	1.58 (0.79)	2.27 (1.62)
Number of seizures in past 6 months	-	1.57 (3.73)	0.00 (0.00)	3.27 (4.94)

IGE split into seizure-free ≥ 6 months (IGE-) and ≥ 1 seizure in 6 months (IGE+)



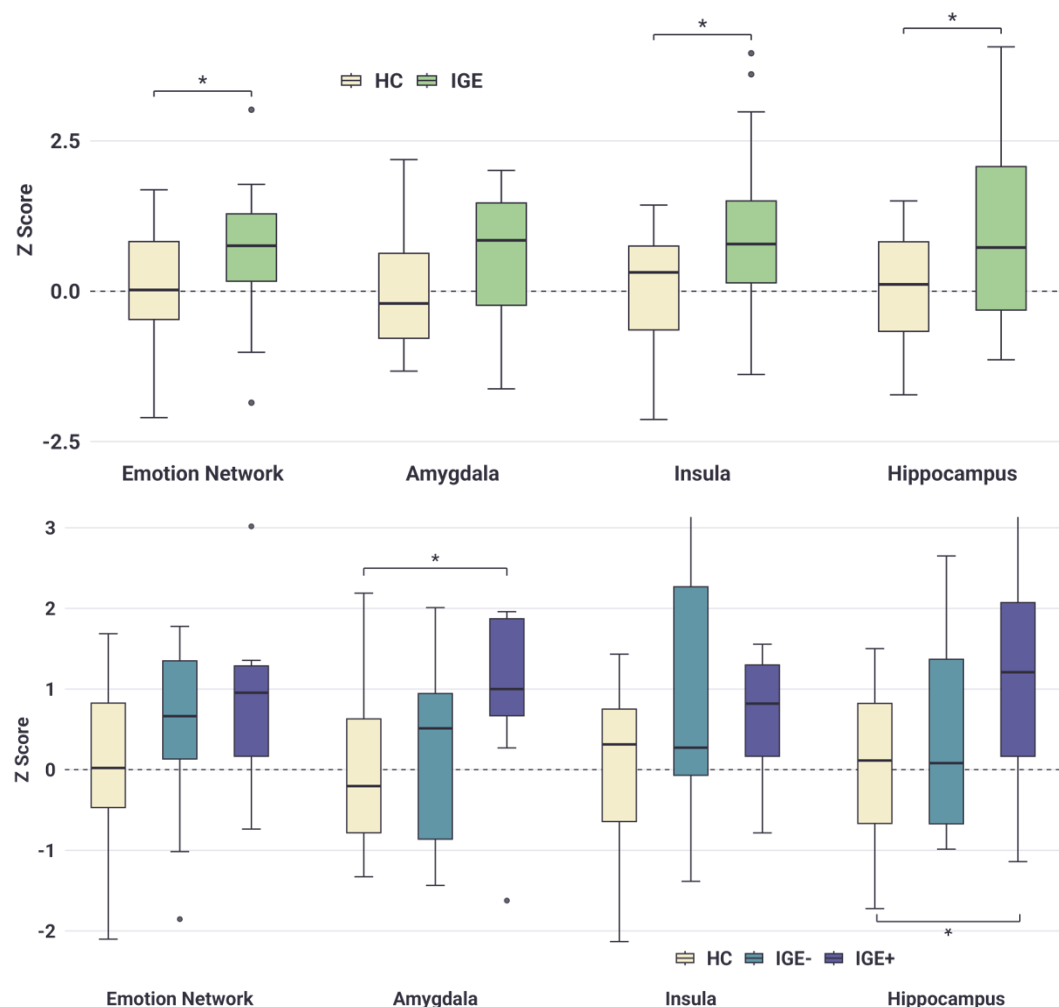
- CBF measured within an emotion network (prefrontal cortex, insula, hippocampus, amygdala)
- Profile of Mood States, Total Mood Disturbance Score

Mood disturbance was substantially higher in IGE, whether or not patients had recent seizures



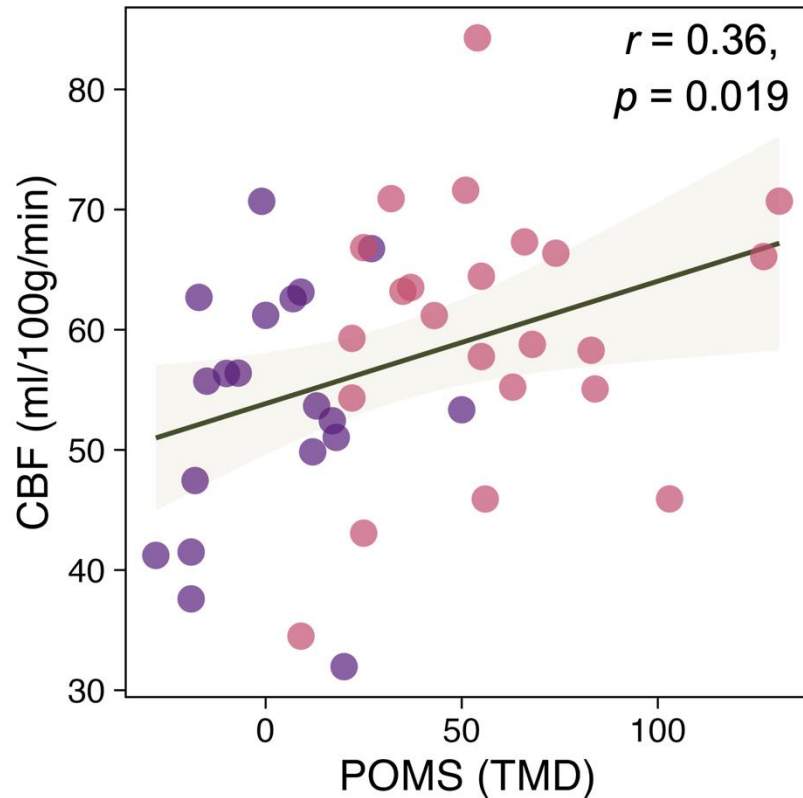
- Total mood disturbance: 57.4 in IGE vs 2.1 in controls ($p < .001$)
- Every POMS subscale was worse in IGE: tension, depression, anger, confusion, fatigue, and vigor
- IGE+ and IGE- did not differ on any POMS score

Emotion-network CBF was higher in IGE, though the effect was modest



- Emotion-network CBF: 60.2 vs 53.5 mL/100 g/min ($p = 0.046$)
- Hippocampus and insula were higher in IGE; amygdala was not
- No significant effect of seizure control on emotion-network CBF
- Hippocampal and amygdala CBF were higher in IGE+ than in controls

Higher CBF was associated with worse mood across the combined sample, but not within either group



Purple: HC; Red: IGE

- Emotion-network CBF correlated positively with total mood disturbance across patients and controls combined ($r = 0.36$, $p = 0.019$)
- Within each group, the correlation was not significant (controls: $r = 0.19$, $p = 0.436$; IGE: $r = 0.22$, $p = 0.304$)
- Small samples and limited within-group variability may explain this difference

What this study can & cannot support

Can support:

- A group-level association between higher emotion-network CBF and worse mood, across patients with IGE and controls combined

Cannot support:

- Causality, direction of effect, or any individual patient inference from CBF

Constraints:

- Small sample (23 IGE, 19 controls)
- Narrow seizure-frequency range
- No formal psychiatric diagnoses
- Heterogeneous ASM regimens
- Lamotrigine reduced CBF in IGE (Joo et al., Eur J Nucl Med Mol Imaging (2006))

Implications & Open Questions

Practice-relevant point:

Mood screening in IGE

Questions for you:

What would it take for an imaging marker to be useful in your clinic?

Are there patients where seizure freedom masks ongoing mood burden in your experience?